

Choroidal melanoma responses after brachytherapy treatment

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Abstract. Serial ultrasonic measurements performed on 85 melanoma patients who were treated with radioactive plaques and followed for up to 141 months revealed that no 2 patients had identical patterns of change. Eighty-two of the patients were categorized into one of four patterns after treatment: Type D (decrease in height, 57 patients), Type S (same height, 12 patients), Type I (increase in height, 9 patients), and Type RG (tumour regrowth, 4 patients). The calculated mean total change in tumor height for Type D was 1.92 mm at 6 months, 2.58 mm at 12 months, 2.97 mm at 24 months and 3.61 mm at 48 months. For Type I, the mean total change in tumor height was 0.80 mm at 6 months, 0.95 mm at 12 months, 2.28 mm at 24 months, and 3.0 mm at 48 months.

Key words: choroidal melanoma - Iodine-125 - Cobalt-60 - radioactive plaque - brachytherapy - tumor of the choroid - tumor response after radiation.

Brachytherapy (Cobalt-60, Iodine-125, Iridium-192, and Ruthenium-106) has been used for treating choroidal melanomas since the 1930's when Stallard developed the vision-preserving technique of suturing rings of Cobalt-60 onto the sclera overlying the tumor of the eye. Although Stallard (1966) reported that the majority of tumors he treated became flat, few other physicians have reported these results. For example, Cruess et al. (1984) reported that choroidal melanomas treated with Cobalt-60 did not become flat, and that regression was not correlated with a patients's propensity for developing metastatic disease. Al-

though Cobalt and Iodine plaques have been used in the United States since 1969 and 1977, respectively, little information is available regarding the response of choroidal melanomas after radioactive plaque treatment.

Material and Methods

A retrospective chart review was performed on 85 patients who were diagnosed with choroidal melanoma and treated with either a Cobalt-60 or an Iodine-125 plaque. All patients received 100 Gy to the apex of the tumor over 7 to 21 days. The diagnosis of melanoma was made clinically. Patients who were previously treated (with treatments such as laser or who had a fine needle biopsy performed) for choroidal melanoma were excluded from the study.

Ultrasound was used to measure the height of the tumor, to estimate the tumor's basal diameter, to evaluate the tumor's proximity to anatomical structures within the eye, and to check for tumor regrowth after treatment. Heights were plotted at monthly intervals and when the patient was not measured during one particular month a mathematically weighted average was taken of the two intervening visits and was plotted accordingly. A change in the tumor height was defined as any measurement greater or less than 15% of the original tumor height measurement.

Results

Of the 85 patients, 54 were male and 31 were female. Of the eyes affected with choroidal melanoma, 52 were right eyes and 33 were left eyes. Location of the tumor vertex was as follows: 54 were posterior to the equator, 8 were at the equator, 14 were anterior to the equator, and 9 were located in the ciliary body. Tumors were also categorized as small, or those less than 3.0 mm in height (5 patients), medium or those measuring 3.1 to 8.0 mm in height (64 patients), and large, or those tumors greater than or equal to 8.1 mm in height (16 patients). Cobalt-60 was used to treat 54 patients, while Iodine-125 was used to treat 31 patients.

The most striking finding from this study is that there is no 'average response' of a treated tumor. No 2 patients had the exact same pattern of response. The magnitude, time course, and direction of change in height was unique for each tumor. All previous studies have averaged the tumor responses and therefore have given the reader the impression that there was a 'common' response which never occurred in any single patient.

It was possible to group 82 of the tumor responses into one of four categories: Type D (for Decrease in height, 57 patients), Type S (for Same height, 12 patients), Type I (for Increase in height, 9 patients) and Type RG (for tumor Regrowth, 4 patients). Three patients did not fit into any of the above four categories because they did not meet the defined criteria.

Type D, or the most common pattern, was defined as those tumors which showed a decrease of at least 15% of the original height after plaque therapy. Patients who had tumors that remained the same height initially but decrease in height at a later time were also included in this group. The mean total change in tumor height for Type D was 1.92 mm at 6 months, 2.58 mm at 12 months, 2.97 mm at 24 months, and 3.61 mm at 48 months.

Type S, or those tumors which remained the same height (or less than a 15% change in height) was the second most common pattern. Type I, included those tumors which showed a 15% increase from the original height beginning at the one month post-operative visit. The calculated mean total change in tumor height for Type I was 0.80 mm at 6 months, 0.95 mm at 12 months, 2.28 at 24 months, and 3.0 at 48 months. Finally, Type RG, the least common type, included those tumors

which initially remained the same (24 months after treatment had less than a 15% change in height) but later regrew more than 15% of the original tumor height.

Four patients had tumors that continued to grow despite the radiation treatment (Type RG). It is noteworthy that regrowth occurred at 26, 29, 47, and 59 months, respectively, and that in only one case did the intraocular tumor increase in height as simultaneous extrascleral extension occurred.

The time course for changes was also different within the four different patterns. For example, tumors were found to decrease in height within one month after treatment. In some cases, the diminution continued for 48 months, while in others, it shrank and remained the same for 11 months.

The diminution was not always linear; the tumors sometimes shrank in spurts. For example, one patient's tumor shrank from 15 mm to 12 mm in 3 months, then remained the same height for 7 months, and then shrank again over a 6-month period. Another tumor shrank 5 mm in the first 2 months after treatment, remained the same height for 12 months, shrank another 2 mm over the next 35 months and then remained the same for 81 months. While it is generally true that shrinkage stops after 48 months, the reader is cautioned that in some cases, shrinkage may terminate from after 12 months after 48 months.

Discussion

Although radioactive plaques have been used to treat choroidal melanomas for almost 50 years, little information is available regarding what happens to the intraocular tumor after treatment. This paper represents one of the largest series of melanoma patients with long-term follow-up after brachytherapy, in whom multiple, serial ultrasonic measurements of height were done. These results help to explain some of the disparate statements in the literature about plaqued tumors and in addition, address misinterpretations in other previously published reports.

Stallard (1966) reported that 70% of his treated melanomas became flat but he did not describe what happened to the remaining 30%. In the two separate series from Wills Eye Hospital and Cornell University Medical College, no tumors were reported to have grown or to have remained the

same in height after treatment. In a report on the first 100 patients from Wills, Cruess et al. (1984) found that the average tumor shrank to 50% of its original thickness after 54 months. However, they made no mention of any cases that remained the same size or actually grew after treatment. Thus, they either deleted the cases that did not decrease in size or simply lost follow-up on those patients whose tumors grew. However, in an earlier paper of the same 100 patients, Shields et al. (1982) reported that three melanomas did increase in height after plaque therapy. Coleman et al. (1984) reported that no melanoma had grown after treatment but that the average changes in height were reported to be between 26.4% and 53.9% at 18 months for different acoustic tissue types.

This review revealed that there were four patterns of response. The majority, or 71%, decreased in height (Type D). This compares to the original work of Stallard (1966) who reported that 70% of his treated patients had tumors which decreased in size. In the present review, 15% of the tumors remained the same height (measured within $\pm 15\%$ of their original height), while another 15% grew after treatment. Four of the tumors initially stayed the same but later regrew intraocularly (Type RG). To average these individual responses confuses and distorts the reality of what happens rather than simplifying the response.

Although four patterns of response were identified, marked variation in magnitude and time course of changes were observed within these patterns. In the patients whose tumors decreased in height (Type D), an elevation could not be detected ultrasonically in any patient; clinically many of these may be similar to those that Stallard described as 'flat'. It appears that total disappearance of plaqued tumors is very rare even though the remaining mass may not contain any malignant cells. The largest shrinkage documented in the present study was 9.5 mm and the largest per cent change in height was 90%.

Conversely, the largest expansion documented was 6.4 mm or a 200% increase in height. When these averaged per cent changes for Type I were compared to those for Type D, tumors in Type I were 185% of their original height at 45 months whereas those in Type D were 50% of their original height at 45 months (Fig. 1). Despite the fact that a change in tumor height from 10 mm to 8 mm is the same reduction in height as a tumor that shrinks

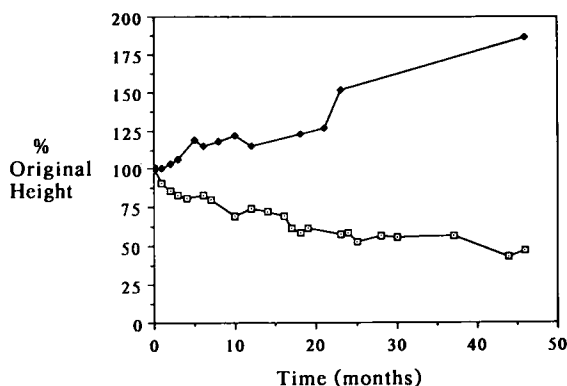


Fig. 1.

Average of percent changes observed in Type I and Type D after plaque treatment over time (months). Type I=solid black squares. Type D=hollow squares with black dots.

from 5 mm to 3 mm, the height changes for the larger tumor represents a greater change in volume because of the cubic function of the radius in volume calculation.

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